

TFPT — Frontier items

η_B , m_p/m_e , the Koide relation, dark matter and quantum gravity —
the honest status, not forced onto the ladder

Stefan Hamann

Alessandro Rizzo

July 25, 2026 · v5.4 (rev 405)

What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of η_B , m_p/m_e , the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
5. **tfpt_5_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: **R.** **tfpt_research_contracts** — the open interfaces (v_{geo} , G_{net} , F_{transfer}) as numbered contracts; **H.** **tfpt_horizon_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** **tfpt_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading document 4 (Frontier).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

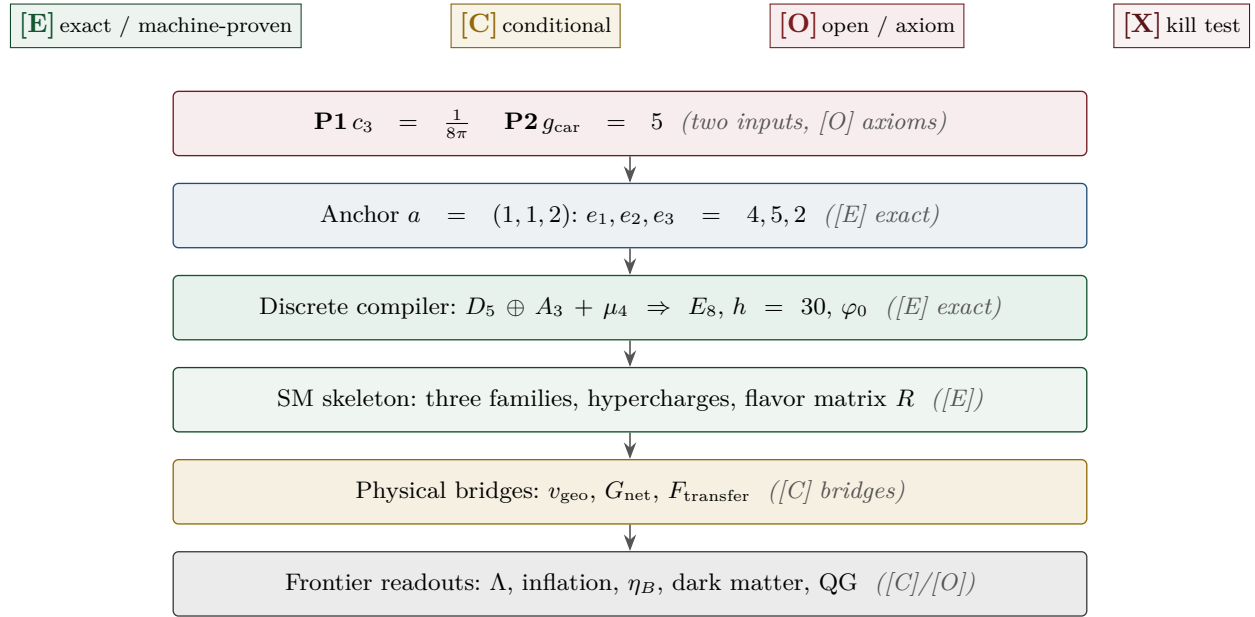
What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B , axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, v358) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit v_{geo} ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= SEAM.EQUIV.01: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems*; the twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, prepared by CELEST.SEAM.01) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (v367/v368) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (v376), the $(E_8)_1$ character with 248 currents and one primary (v377), genus-one torus GSD = 1 (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the #print axioms check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ($h_s=16/16=1 \in \mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM $\mu=4/2^2=1$; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route seamResidualClosed' (v469); so the MMST route is *closed modulo cited theorems*, not solved. QGEO.SYM.01 (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, v76/v330). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of verification/status_ledger.csv (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and QG.AMB.01 a [C] redundancy.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status_ledger.csv.



Contents

1 Baryon asymmetry η_B — a downstream readout from the closed $\Omega_b h^2$	4
2 Proton/electron ratio m_p/m_e — a cross-sector ratio	6
3 The Higgs quartic — near-criticality from the free seam	7
4 The Koide relation — near $2/3$, computed exactly	8
5 Dark matter — the candidate is fixed, the scale is pending	10
6 The muon anomalous magnetic moment — a seam vertex readout	12
7 Full quantum gravity — induced from the boundary kernel	13
Summary	17
Appendix: computational verification	18

Ground rule. These five are exactly the items previously flagged “*not on a clean action/ladder yet — do not force.*” This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers **[E]**/**[C]**/**[O]** are applied throughout. (All numbers machine-verified.)

Hard rule on the frontier — one interface F_{transfer}

Koide, η_B , the axion relic scale and m_p/m_e are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied. They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor* F_{transfer} — the continuous transport from compiler *source* data to measured observables (Koide source→pole; η_B source→Boltzmann relic; axion scale→cosmological abundance; $m_p/m_e \rightarrow$ QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named transfer calculation — now a runnable **[C]** solver with a kill test (v371–v375), still a transfer target, never a compiler power. In particular η_B is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

Why this is principled, not a gap: these are precisely the **discrete**↔ **continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [[verification/v35_quark_qcd_boundary.py](#)]

Machine-enforced. This rule is a CI-style guard ([[verification/v187_ftransfer_laws.py](#)]): the four frontier families (F_{pole} Koide, $F_{\text{Boltzmann}}$ η_B , F_{relic} axion, F_{QCD} m_p/m_e) are read from the ledger and required to carry a **[C]**/**[O]** marker — exact algebraic sub-parts (the Koide 53/54 factor, $b_3 = -7$) may be **[E]**, but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([[verification/v188_frontier_wording_guard.py](#)]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

*The functor contract **CONTRACT.F.01**.* Beyond the typing guard, the functor is pinned by four checkable structural axioms ([[verification/v213_ftransfer_functor.py](#)]): each instance is a *discrete* compiler kernel **[E]** composed with an *external* continuous solver **[C]**, and the functor is (1) μ_4 -deck equivariant (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue), (2) Plücker-preserving ($53 = a^T(R+Q)\mathbf{1}$, $\|\text{Pl}(K)\|_1 = 11$), (3) positive/stochastic ($\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$, $\kappa_f \in (0, 1)$), and (4) external-module-explicit ($b_3 = -7$ a carrier output, the Boltzmann/relic/QCD content named and external). So F_{transfer} is a typed functor, not a bag of open topics — the third research contract alongside (U_{wall}) and (G_{metric}), and a consolidation, not a closure.

How external is “external”? A falsifiable clock probe ([[verification/v397_external_clock_probe.py](#)]). Under the strict No-Free-Pattern rule (an *exact* match, not a wide-band fit), three of the four frontier scheme-numbers in fact land on the $\{2, 3, 5\}$ clock: the η_B decuple $A_\Lambda = 10 = |Z_2| g_{\text{car}}$ (v212), the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ (v303) and the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211). Only the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value (several simple numbers fit its $\pm 5.3\%$ band), so it stays genuinely external — the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number.

The Paper-E closure certificate ([[verification/v402_ftransfer_suite.py](#)], **FTRANSFER.SUITE.01**). The four sectors are certified as ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$: each observable factors $\text{source} \rightarrow \text{threshold} \rightarrow \text{RG} \rightarrow \text{observable}$ from an exact atom source (Q_{src} , $A_\Lambda = 2g_{\text{car}} = 10$, $f_a = M_{\text{scal}}/128$, $b_3 = -7$) and carries a committed kill test, with the firewall intact (the physical prediction stays **[C]**/**[O]**, only the algebraic source **[E]**) — a consolidation of the **FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE** solvers, not a closure.

1 Baryon asymmetry η_B — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

Route A — from the closed baryon fraction. The seed already fixes the baryon *fraction* (the E_8 decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \frac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \quad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed $\Omega_b \approx 0.049$. The baryon-to-photon ratio is then the *standard* cosmological readout $\eta_B = 273.9 \times 10^{-10} \Omega_b h^2$. With the TFPT Hubble value ($h \approx 0.674$, from $H_0 \sim \sqrt{\Lambda}$),

$$\boxed{\Omega_b h^2 = 0.0222, \quad \eta_B = 6.09 \times 10^{-10}} \quad (\text{observed } 6.1 \times 10^{-10}).$$

So η_B is a *downstream cosmological readout* from the closed $\Omega_b h^2$ handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

Route B — leptogenesis interface ([C]**).** The TFPT branch fixes the neutrino spectrum, the CP phase structure (δ_{CKM} , δ_{CP}^ν , both closed) and a *plausible washout anchor* $\tilde{m}_1 = m_3/A_\Lambda \approx 5$

meV. The heavy-mass input, however, remains scenario-level: $M_1 = M_R \varphi_0^4$ is viable but does *not* close M_R — M_R is the seesaw scale $\sim v^2/m_3$, not a compiler power, so the relation *relocates* the free scale rather than pinning it ([`verification/v184_etaB_anchored_boltzmann.py`]). The numerical η_B then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero. **Honest caveat:** that Boltzmann network (the washout/efficiency κ_f , the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed $\Omega_b h^2$) is quoted.

The interface, operationalised ([C]). The Boltzmann path can be made falsifiable: with the standard estimate $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted $\delta_{\text{CP}}^\nu = 4\pi/3$ — the Davidson–Ibarra asymmetry $\varepsilon_1 = \frac{3}{16\pi} M_1 m_3 / v^2$ and the washout efficiency $\kappa_f(\tilde{m}_1)$ give, over the natural ranges $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and $\tilde{m}_1 \in [m_*, m_{\text{atm}}]$, an η_B band $[8.4 \times 10^{-11}, 4.7 \times 10^{-9}]$ that *brackets* the observed 6.1×10^{-10} (a canonical point $M_1 = 10^{10}$ GeV, $\tilde{m}_1 = 5 \times 10^{-3}$ eV gives 6.0×10^{-10} , untuned). So the route is *viable*; but M_1 and the washout are scenario inputs, so η_B stays [C] — if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([`verification/v169_etaB_boltzmann_interface.py`]).

Honest status of η_B — strictly typed [C]

The two statements must be kept apart and not interchanged:

η_B as a cosmological readout from the closed $\Omega_b h^2$ is *determined* (6.1×10^{-10})

η_B as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So η_B is [C] — determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

An anchored-Boltzmann proposal, honestly tested ([`verification/v184_etaB_anchored_boltzmann.py`]).

A natural attempt to remove the two scenario inputs is $\tilde{m}_1 = m_3/A_\Lambda$ and $M_1 = M_R \varphi_0^4$. The first *works* as a [C] anchor: $\tilde{m}_1 = m_3/10 \approx 5 \times 10^{-3}$ eV ($A_\Lambda = 10 = |E(K_5)|$ the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*: $M_R \sim v^2/m_3$ is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of $\bar{M}_{\text{Pl}}$ both miss it (the c_3 - and φ_0 -powers are off by $\sim 75\%/ \sim 40\%$), so $M_1 = M_R \varphi_0^4$ merely *relocates* the free input $M_1 \rightarrow M_R$. Feeding both into the standard estimate lands η_B in the observed band $O(10^{-10})$, so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero. η_B **stays [C], a sharper scenario, not a derivation.**

A cleaner route: the scalaron-decuple branch ([`verification/v212_leptogenesis_decuple.py`]). Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$:

$$M_1 = \frac{M_{\text{scal}} (\varphi_0^{\text{ret}})^2}{A_\Lambda} \approx 8.65 \times 10^9 \text{ GeV}, \quad \tilde{m}_1 = \frac{m_3}{A_\Lambda} \approx 5 \text{ meV}.$$

M_1 now uses *only* compiler quantities ($M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$, φ_0^{ret} , A_Λ) — *no hidden seesaw scale* — and lands inside the thermal window $[3 \times 10^9, 3 \times 10^{10}]$ GeV where the Davidson–Ibarra $\varepsilon_1 \approx 8.5 \times 10^{-7}$ and the strong-washout efficiency give $\eta_B \approx 1.2 \times 10^{-9}$, bracketing the observed 6.1×10^{-10} untuned. The *full* BDP Boltzmann ODE (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`) now confirms this: the integrated efficiency $\kappa_f = 0.092$ (validating the analytic κ_f) gives $\eta_B = 6.5 \times 10^{-10}$ at the frozen M_1 — a factor 1.07 from the observed value, with *no* free M_R dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing A_Λ reads as *scalaron reheating dressed by two seed insertions* $(\varphi_0^{\text{ret}})^2$, over the pair sector. This is cleaner than

$M_1 = M_R \varphi_0^4$ (no relocated free scale), but the coupling *mechanism* is posited, not derived, and κ_f is washout-dependent — so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

2 Proton/electron ratio m_p/m_e — a cross-sector ratio

m_e is *closed* on the φ_0^{ret} -ladder ($\hat{m}_e = \frac{v_{\text{geo}}\pi}{\sqrt{2}} \frac{16}{7} (\varphi_0^{\text{ret}})^5$). The proton mass is *not* a Yukawa: it is QCD-dominated, $m_p \sim$ a few $\times \Lambda_{\text{QCD}}$, and Λ_{QCD} arises by dimensional transmutation from α_s running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single φ_0^{ret} power: $1/(\varphi_0^{\text{ret}})^2 = 353.7$ and $1/(\varphi_0^{\text{ret}})^3 = 6651$ bracket 1836 with no clean rung in between.

The numerator is parameter-free, the ratio is not forced. Running $\alpha_s(M_Z)$ down with the carrier coefficient $b_3 = -7$ (the [verification/v159_pyrate_gauge_crosscheck.py] value, two-loop + n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and reaches confinement ($\alpha_s \sim 1$) at ~ 0.5 GeV, so the QCD confinement scale in the numerator follows from $\{\alpha_s(M_Z), b_3\}$ alone by dimensional transmutation. The $O(1)$ proton/ Λ factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([verification/v164_qcd_scale.py]).

The F_{QCD} transfer is now an implemented solver ([verification/v262_fqcd_mp_me.py]). The contract-only pipeline (experiments/ftransfer/qcd_matching_mp_me) is realised: a two-loop α_s run from M_Z with n_f thresholds (carrier coefficient $b_3 = -7$, [E]) gives $\alpha_s(2 \text{ GeV}) \simeq 0.30$ and $\Lambda_{\overline{\text{MS}}}^{(3)} \simeq 0.33$ GeV; with the lattice bridge $C_p = m_p/\Lambda^{(3)}$ (external $O(1)$) and the closed m_e this reproduces $m_p/m_e \simeq 1824$ against the measured 1836.15 within the external error budget ($\sim 7\%$, lattice- C_p dominated). This advances FR.MPME.01 ([O]) to FR.MPME.02 ([C], “reproduced by the named F_{QCD} transfer”); the firewall holds — $b_3 = -7$ is [E], the ratio is a transfer, never a compiler power.

Honest status of m_p/m_e [O]

Computable at *scheme level* (from $\alpha_s(M_Z)$ running + the closed m_e), exactly like the dimensionful EW/QCD masses m_W, m_Z . But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

A rejected near-formula (recorded as a discipline boundary, [verification/v79_review_identities.py]). The combination $|W(D_5)| - \dim \Lambda^3(SU(9)) + N_{\text{fam}} \lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$ matches the PDG value to $\sim 10^{-6}$ — and is *deliberately rejected*. Two reasons make it numerology, not a derivation: $SU(9) = A_8$ (the $\dim \Lambda^3 = 84$ block) does *not* occur in the carrier $D_5 \oplus A_3$, and m_p is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it; m_p/m_e stays [O]. (The same firewall rejects $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q} c_3^6 \approx 6.10 \times 10^{-10}$: arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output — η_B stays [C], the downstream readout above.)

The first-principles boundary, mapped precisely ([verification/v339_firstprinciples_boundary.py]).

Two questions are worth answering head-on: can the Planck mass v_{geo} be *computed*, and can F_{transfer} be made a first-principles compiler output? Both answers are *no* — and the residuals are not the same kind of gap. They classify into four honest kinds. [E] $v_{\text{geo}}/M_{\text{Planck}}$ is *forbidden by theorem* (No-Unit, v153): a mass carries mass-dimension while every TFPT datum is dimension-0, so no dimensionless map can output an absolute scale; it is not a gap but is over-determined (gravity = dark energy to 0.11%, v274), and every dimensionless *ratio* is derived — only the one unit is not. [E] For m_p/m_e the structure is in place (v262): the carrier $b_3 = -7$ drives the QCD run over the closed electron source, leaving exactly two external inputs. [X] One of them, $\alpha_s(M_Z)$, is external *because of a tension*: the SM (= the TFPT gauge content, no new states) does not

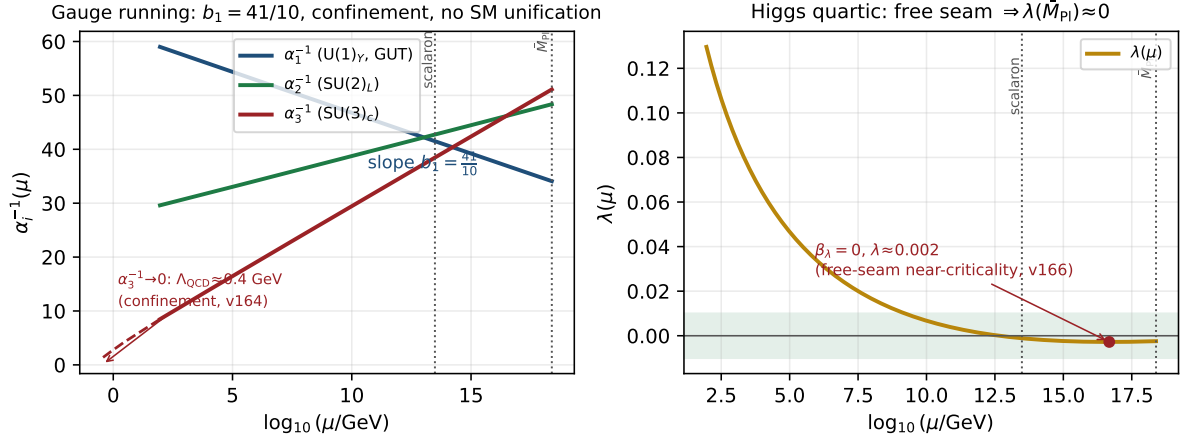


Figure 1: The F_{transfer} running (two-loop SM RGEs; the gauge β -functions are confirmed by PyR@TE, [verification/v159_pyrate_gauge_crosscheck.py]). *Left*: the gauge couplings — the $U(1)_Y$ slope is $b_1 = 41/10$, $\alpha_3^{-1} \rightarrow 0$ at $\Lambda_{\text{QCD}} \approx 0.4$ GeV (confinement, [verification/v164_qcd_scale.py]), and the Standard Model does *not* unify. *Right*: the Higgs quartic falls to $\lambda(M_{\text{Pl}}) \approx 0.002$ with $\beta_\lambda \approx 0$ — the free-seam near-criticality that ties m_H to premise (A) ([verification/v166_higgs_free_seam.py]).

unify (v246), so no unified coupling is predicted. [C] The other, the lattice $C_p = m_p/\Lambda^{(3)} \sim 2.83$, is parameter-free but lattice-only (the discrete $\leftrightarrow$ continuous handoff, v35); and η_B /axion ride on cosmological initial conditions. So F_{transfer} cannot be made first-principles in the current theory — not for lack of a formula, but because the residuals are, respectively, forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, and cosmological. An honest boundary, stated, not a solution.

3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral $c=8$ fixed point ([verification/v158_fixed_point_stable.py]): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling λ is the one marginal scalar coupling, so a free seam forces it to the Gaussian point there:

$$\lambda(M_{\text{seam}}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\text{seam}}) = 0$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from M_Z upward with the measured (m_H, m_t) gives $\lambda(M_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(M_{\text{Pl}}) \approx 0$ at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate ~ 0.022). Imposing the double condition predicts $m_H \sim 129\text{--}134$ GeV (a band, sensitive to m_t and α_s); the measured 125.25 GeV sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ($c_3^{7/2} M_{\text{Pl}} \sim 3 \times 10^{13}$ GeV) gives $m_H \sim 107$ GeV (too low), so the boundary condition lives at M_{Pl} , consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only $N_\Phi = 1$); it is typed [E] (the freeness signature and scale selection) and [C] (the m_H band) ([verification/v166_higgs_free_seam.py]). Figure 1 shows the whole F_{transfer} running.

4 The Koide relation — near $2/3$, computed exactly

With the closed lepton source masses $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\text{ret}})^5, \frac{4}{3}(\varphi_0^{\text{ret}})^3, \frac{7}{6}(\varphi_0^{\text{ret}})^2)$ the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\text{TFPT}} = \frac{\sum_\ell \hat{m}_\ell}{(\sum_\ell \sqrt{\hat{m}_\ell})^2} = 0.66446\dots, \quad \frac{2}{3} = 0.66667, \quad \text{deviation} = -0.33\%.$$

The measured pole-mass value is famously $Q_{\text{PDG}} = 0.66666$. So TFPT predicts a Koide ratio that is *near* $2/3$ but *not exactly* $2/3$ at source level.

Honest status of Koide [C]

$Q_{\text{TFPT}} = 0.664$ is a *prediction*, 0.33% below $2/3$. Exact $2/3$ would require the three carrier rationals $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force $2/3$.

The *target* $2/3$ has a compiler reading $|\mathbb{Z}_2|/N_{\text{fam}}$. The E_8 -slice scan adds the missing piece: the value the data sits on, $Q = 2/3$, is exactly $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$, the sheet over the family count — the same two atoms $(2, 3)$ that run through the whole compiler. So the picture splits cleanly: the *democratic target* is $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus “democratic $2/3$ from $(|\mathbb{Z}_2|, N_{\text{fam}})$, realised to 0.33% by the explicit masses” — promoted from a bare near-miss to a reading with a target. [C]

Conjecture: a source→pole correction of size $\varphi_0/|W(A_3)|$ ([C]). The source value sits just below $2/3$, and the deficit is almost exactly the family-Weyl quantum $\varphi_0/|W(A_3)|$ with $|W(A_3)| = 4! = 24$:

$$Q_\ell^{\text{source}} = 0.6644638, \quad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting $\frac{2}{3} = 0.6666667$ by only 1.3×10^{-5} . So the *conjecture* $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$ would land essentially on the democratic $2/3$. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [verification/v25_frontier_conjectures.py]

Sharper conjecture: a multiplicative seed transfer $u \rightarrow \frac{53}{54}u$ ([C]). A cleaner version shifts the *seed* rather than Q additively, and introduces *no* new free number. Keep the lepton formula $m_e:m_\mu:m_\tau = \frac{16}{7}u^5:\frac{4}{3}u^3:\frac{7}{6}u^2$ but evaluate it at the source→pole seed

$$u_{\text{pole}}^{(\ell)} = \left(1 - \frac{1}{2N_{\text{fam}}^3}\right)u = \frac{53}{54}u, \quad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2| N_{\text{fam}}^3.$$

Then $Q_\ell(\frac{53}{54}u) = 0.66666610$, i.e. $2/3$ to within -5.7×10^{-7} — an order of magnitude tighter than the additive $\varphi_0/24$ correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is $u \mapsto u(1 - \frac{1}{2N_{\text{fam}}^3})$* (a renormalised-seed effect from the leptonic self-energy or the A_3 family-Weyl averaging). Status remains [C] until that transfer is derived. [verification/v37_plucker_anchor.py]

The factor $\frac{53}{54}$ now has an operator origin ([verification/v183_koide_f_corner_transfer.py]). It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner $F = R + Q = M(1, 1)$ over the sheet-doubled $E_6 \times A_2$ cubic family block,

$$\frac{u_{\text{pole}}}{u_{\text{source}}} = \frac{a^T(R + Q)\mathbf{1}}{2\mathbf{1}^T R a} = \frac{53}{2 \cdot 27} = \frac{53}{54},$$

where $\mathbf{1}^T R a = 27$ is the cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$) and $a^T(R + Q)\mathbf{1} = 53 = 52 + 1$ with $\det B_F = 52 = \dim F_4$ (v80). A hard negative control fixes the corner: $a^T M \mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. So the

factor is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the F corner — an F_{transfer} special case (F_{pole}). Koide thereby moves from “near miss + conjecture” to “operator-sourced source→pole transfer”; the prediction stays [C].

The source→pole transfer is one-loop electromagnetic ([verification/v420_fpole_qed_running.py]).

Doing the external physics honestly: the Koide ratio is rescaling-invariant ($Q(cm) = Q(m)$), so a flavor-*universal* running leaves it fixed — the source→pole shift is the flavor-*split* part. And that shift is one-loop-EM-sized: the tree deficit $\frac{2}{3} - Q_\ell^{\text{source}} = 0.00220$ equals the seed quantum $\varphi_0/24$ to 0.6% and the electromagnetic scale α/π to $\sim 5\%$, with the pole sitting at $2/3$ (the IR attractor). A 1-loop QED pole↔ $\overline{MS}$ estimate (per-lepton log) moves Q by an EM-loop-order amount toward $2/3$. The *exact* flow time ($t \approx 2.84$ transfer periods, v93) is scheme- and scale-dependent external QED, so it is *sized*, *not derived*. This identifies F_{pole} as a one-loop EM effect — the algebraic lens forces the *rate* $(2/3)^6$ (v327), this external physics supplies the *flow* — and Koide stays [C], firewalled (v187).

The Koide target and the QFT transfer gap are the same quotient ([E]/[C]). The same $Q_\star$ controls a second, independent object — the first non-trivial eigenvalue of the admissible C_6 transport operator. Its spectrum is $\{1, (2/3)^6, (1/3)^6\}$, so

$$\lambda_2 = \left(\frac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \quad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6.$$

The Koide democratic target, the QFT mass-gap eigenvalue λ_2 (hence $\Delta = 6 \log \frac{3}{2} = -\log \lambda_2$), and the A_3 positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not “solve” Koide, but it explains *why* the near-value $2/3$ recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

The source→pole map is forced by that gap ([E]/[C]). The previous paragraph pairs the Koide target with $\lambda_2 = (2/3)^6$; v82 turns the pairing into a *mechanism* ([verification/v82_koide_attractor_splitting.py]). On the anchor double cover (the flavor sector) the two branch points are Koide $q = 2$ and carrier $q = 5$; any branch-preserving Möbius transfer that fixes both is, in $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$, hence *unique* once μ is set. Choosing $\mu = \lambda_2 = (2/3)^6$ (no new number) gives the unique map $F(q) = 2(569q - 3325)/(665q - 3517)$ with $F'(2) = (2/3)^6 < 1$ and $F'(5) = (3/2)^6 > 1$: **Koide is the attractor, the carrier the repeller**, and $q_n \rightarrow 2$ ($x \rightarrow -2/3$) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value $2/3$, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture $u \mapsto (53/54)u$ above; both contract the source onto $2/3$.)

Relaxation toy: basin, exact rate, and two honest negatives ([E]). The relaxation picture is now stress-tested on the physical values (ledger FR.KOIDE.05, [verification/v93_koide_relaxation_toy.py]).

Basin lemma [E] (new): under the canonical dictionary $q = N_{\text{fam}} Q$ the *entire* physical Koide range $Q \in [\frac{1}{3}, 1]$ (Cauchy–Schwarz) maps to $q \in [1, 3]$ — the attractor $q = 2$ lies inside, the repeller $q = 5$ outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly* $(2/3)^6$ per step. **Source/pole [E]:** the exact ladder gives $Q_{\text{src}} = 0.6644638$, i.e. $\rho_{\text{src}} = -0.0021980 \approx -\varphi_0/24$ to 0.8% — the $\varphi_0/24$ conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ($24 = |W(A_3)|$); PDG pole masses give $Q_{\text{pole}} = 0.6666645$, the branch point hit to 3.3×10^{-6} . **Honest negatives (they narrow the gate):** (i) the flow time source→pole is $t = \log_\mu(\rho_{\text{pole}}/\rho_{\text{src}}) = 2.84$ — *not* an integer, so the literal discrete iteration $\text{pole} = F^n(\text{source})$ is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator $\exp(t \log F)$; (ii) the 0.79%

mismatch of ρ_{src} vs $-\varphi_0/24$ is far above ladder precision, so “exactly $\varphi_0/24$ ” remains a conjecture, not an identity. The open question is no longer “why $2/3$?” but “what is the continuous transfer generator, and why is its flow time ~ 2.8 periods?”

Flow time: canonical generator, data-fragility, and a conditional m_τ ([E]/[C]). Three sharpenings of the paragraph above (ledger FR.KOIDE.06, [verification/v99_koide_flow_time.py]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\text{fam}}} (q-2)(q-5) = \frac{\Delta}{N_{\text{fam}}} \det B(q)}, \quad \Delta = 6 \log \frac{3}{2} \quad (\text{the gap}),$$

whose time-1 map *is* the Möbius map F (machine-checked; $e^{-\Delta} = (2/3)^6$ exactly). The missing [C] object is therefore precisely a QFT mechanism producing $\text{gap} \times \text{anchor-block quadric}$ — the form is canonical, only its physical derivation is open. **(2) The non-integrality of t is data-fragile [E].** Because $\rho_{\text{pole}} = -2.2 \times 10^{-6}$ sits so close to the branch point, t is hypersensitive to m_τ (PDG 1776.93 ± 0.09 MeV): $t(-1\sigma) = 2.35$, $t(\text{central}) = 2.84$, and Q crosses $\frac{2}{3}$ *inside* the error band (at $+0.43\sigma$, where $t \rightarrow \infty$) — within $+0.5\sigma$, t passes *every* integer ≥ 3 . So the exclusion of pole = $F^n(\text{source})$ holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):** $n=2$ is *excluded* (-2.9σ , [E]); $n=3=N_{\text{fam}}$ steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427 \text{ MeV}} \quad (+0.14\sigma; \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while $n=4$ (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is $\sigma(m_\tau) \sim 0.01$ MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading $t = \ln(M_{\text{scal}}/m_\tau)/\ln W$ matched $W = 46080 = |W(D_5)||W(A_3)|$ to 10^{-4} — and is *destroyed* by negative controls (the same hypersensitivity makes *any* W in a wide range land within 0.1σ); it is rejected and recorded so it is not re-fished. P2’s type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Its *candidate identity* is fixed by the E_8 scan (below) and the domain-wall number N_{DW} winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\text{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\text{str}}(f_a, N_{\text{DW}}).$$

The initial misalignment angle is *not* a single closed number: the finite- T misalignment solve (below) selects the *spine* branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5 = 108^\circ$ as the robust dominant-DM value in the *pre-inflationary* reading (it lands $\Omega_a h^2$ on Ω_{DM} untuned, in the mild-anharmonic regime; but superseded by the post-inflationary reframing below), while the seam hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}(\alpha_*)) = 170.4^\circ$ *over-produces* ($\sim 5.4\times$) and is disfavoured. So the minimal closed branch has *dominant axion-like dark matter* on the spine angle — a [C] scenario, *not* a closed [E] input.

The E_8 scan rules out a rival WIMP candidate. The $D_8 = SO(16)$ slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor $128 = (16, 4) \oplus (\overline{16}, \overline{4})$ is the *same* matter spinor as in the $D_5 \times A_3$ construction — there is *no spare E_8 singlet* left over to play a WIMP. Every E_8 direction is accounted for by the carrier $\times$ family content. So dark matter cannot be a new E_8 representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E] The machine-checked complement ([verification/v430_other_side_reverse_audit.py]) makes this explicit: the spinor $128 = 2 \cdot 64 = \sum \deg$ and the double-cover deck $|\mathbb{Z}_2| = 2$ lie in E_8 ’s *matched* structure

and are forced-disjoint from the five unmapped Casimir degrees $\{12, 14, 18, 20, 24\}$, so the seam's "other side" hosts neither a spare singlet nor the unmapped hull overhead.

Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion, [E]: the E_8 scan leaves no spare singlet for a WIMP). The misalignment angle is a [C] *branch decision*, not a closed [E] number: the finite- T solve selects the spine $\theta_i = 3\pi/5 = 108^\circ$ (untuned, in band) over the over-producing 170° hilltop. The relic abundance still needs the decay constant f_a and mass m_a (interface data set by the M_R/Λ scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate fixed, robust spine angle, scale f_a pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near M_R .)

Conjecture: a determinant-line axion scale $f_a = M_{\text{scal}}/128$ ([C]). A natural compiler scale for the decay constant is the scalaron mass over the full carrier $\times$ deck multiplicity,

$$f_a = \frac{M_{\text{scal}}}{2 \dim S^+ |\mu_4|} = \frac{M_{\text{scal}}}{128}, \quad M_{\text{scal}} = c_3^{7/2} M_{\text{Pl}} \approx 3.06 \times 10^{13} \text{ GeV},$$

giving $f_a \approx 2.39 \times 10^{11} \text{ GeV}$ and, via the standard QCD-axion relation $m_a \simeq 5.70 \mu\text{eV} (10^{12} \text{ GeV}/f_a)$, a mass $m_a \approx 23.8 \mu\text{eV}$. With the robust spine angle $\theta_i = 3\pi/5 = 108^\circ$ (below) this f_a lands the relic on Ω_{DM} without hill-top tuning; it remains a *cosmological scenario*, not a closed hit. [verification/v25_frontier_conjectures.py]

The haloscope coupling is tied to c_3 . In the determinant-line normalization (Lagrangian $\mathcal{L} \supset c a F \tilde{F}$) the axion-photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \quad y^2 \equiv g_{a\gamma\gamma}^2 = 16c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and the cosmic birefringence $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$ in GeV^{-1} , which still carries f_a). The small-angle reading $\theta_i = \varphi_0$ of the determinant-line axion (an earlier note) is *superseded* by the robust spine angle $\theta_i = 3\pi/5 = 108^\circ$ selected by the finite- T solve below (the seam hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ over-produces); both earlier readings are recorded as retired, never used.

Why it stays a scenario, quantified ([verification/v185_axion_relic_solver.py]). The harmonic misalignment *under-produces* at this f_a : $\Omega_a h^2 \sim 0.12 (f_a/9 \times 10^{11})^{7/6} \langle \theta^2 \rangle$ gives a prefactor ~ 0.026 per θ^2 , so $\theta_i \sim O(1)$ would make only $\sim 21\%$ of Ω_{DM} ; reaching the observed $\Omega_{\text{DM}} h^2 = 0.12$ *requires* $\theta_{\text{eff}}^2 \sim 4.7$, i.e. exactly the large-angle (anharmonic) enhancement at the $\theta_i \approx 170^\circ$ hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to θ_i and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract $F_{\text{relic}}(f_a=M_{\text{scal}}/128, \theta_i \approx 170^\circ, N_{\text{DW}}=1) \stackrel{?}{=} \Omega_{\text{DM}}$ is a kill test for the *branch*, not the theory.

A full finite- T solve confirms an over-production tension. Both a quick $\theta^2 F$ estimate ($F(\theta_i) \approx 4$ at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + \left(3 + \frac{d \ln H}{dN}\right) \theta' + (m_a(T)/H)^2 \sin \theta = 0$$

(axion_relic/full_finiteT_solve.py: lattice susceptibility $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$, converged and normalised so $\theta_i=1$ gives the standard $\Omega_a h^2 \approx 0.03$) agree: at the *predicted* $\theta_i \approx 170^\circ$ the relic is $\Omega_a h^2 \approx 0.66$, $\sim 5.5\times$ above $\Omega_{\text{DM}} h^2 = 0.12$. The observed value is reached only at $\theta_i \approx 106^\circ$, not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at $(f_a=M_{\text{scal}}/128, \theta_i \approx 170^\circ)$ *over-closes* the universe unless there is extra dilution (entropy injection) or a lower f_a . This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

A more robust angle: the spine branch. [verification/v211_axion_spine_angle.py] The same finite- T solver reaches Ω_{DM} at $\theta_i \approx 106^\circ$ — and the central spine quotient gives $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5 = 108^\circ$ *exactly* (no fit; harmonic estimate $0.03\theta_i^2 = 0.107$, lifted to ~ 0.12 by a mild anharmonic factor). [verification/v429_axion_pentagon_phi.py] This angle is not a bare rational multiple of π : because $N_{\text{fam}} = g_{\text{car}} - 2$, $\theta_i = (g_{\text{car}} - 2)\pi/g_{\text{car}}$ is *exactly* the interior angle of the regular g_{car} -gon — for $g_{\text{car}} = 5$ the *pentagon*, so $\cos\theta_i = \cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$ with $\varphi = 2\cos(\pi/5)$, and this golden character is *unique* to $g_{\text{car}} = 5$ (the small n -gons $n = 3, 4, 6$ give the rational $\frac{1}{2}, 0, -\frac{1}{2}$). So the otherwise-unmapped golden/icosahedral $g_{\text{car}} = 5$ signature (v354/v313) is the geometry of this one external input — a [C] reading that does *not* upgrade DM.AXION.SPINE.01. Crucially 108° lies 62° *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to $\theta_i = \pi(1 - \varphi_\Sigma) \approx 170^\circ$ (the two are mutually exclusive; neither is forced) — a sharper [C] scenario decided by the full solver (DM.AXION.SPINE.01), *not* a derivation; [X] a converged $\Omega_a h^2$ outside $\sim [0.08, 0.16]$ at $\theta_i = 3\pi/5$ demotes the branch.

Isocurvature and the post-inflationary reframing ([C]). Both fixed-angle readings above are *pre-inflationary*: they assume Peccei–Quinn breaking *during* inflation with a homogeneous θ_i . But the R^2 branch fixes $H_{\text{inf}} = \pi M_{\text{Pl}} \sqrt{A_s r}/2 \approx 1.6 \times 10^{13}$ GeV, so $H_{\text{inf}}/f_a \approx 66 > 1$ and $H_{\text{inf}}/(2\pi f_a) \approx 10$: the de Sitter fluctuation of the angle exceeds its range, the symmetry is effectively *restored* during inflation, and a fixed pre-inflationary θ_i (either 108° or 170°) would carry CDM isocurvature far above the Planck bound $\beta_{\text{iso}} < 0.038$. The clean, *dial-free* resolution is the *post-inflationary* scenario: PQ breaks after inflation, θ_i is a horizon average (*not* a prediction), and the relic follows from f_a alone — with $f_a \approx 2.39 \times 10^{11}$ GeV in the post-inflationary window ($\Omega_a h^2 \sim O(0.1\text{--}0.5)$). It is domain-wall-safe because the determinant-line axion’s colour anomaly is the seam Dirac index = the inflow level $|C| = 1$ (v470/v472), so $N_{\text{DW}} = 1$ (a [C] identification). Thus the spine/hilltop *angle* is not an observable and the branch is *reframed*, not killed: drop the fixed angle, keep f_a and $N_{\text{DW}} = 1$, and the relic is a robust post-inflationary prediction. [X] a post-inflationary relic outside the plausible window at this f_a (with $N_{\text{DW}} = 1$) demotes the branch. (Strategy scan and the N_{DW} computation: experiments/theory-contracts/axion01-06.)

6 The muon anomalous magnetic moment — a seam vertex read-out

A [C] downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes α : with $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4 = 48c_3^4 = 3/(256\pi^4)$ and the carrier compression quotient $B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}$ ([verification/v51_boundary_half_step.py]), $\delta_2 = B\gamma \delta_{\text{top}}^2 = \frac{5}{4} \delta_{\text{top}}^2$. Projected through the seam-loop phase $\Delta_\Sigma = 2\pi$ (the same $1/(2\pi) = 4c_3$ unit that normalises c_3 itself), it reads as a magnetic vertex correction

$$a_\mu^{\text{seam}} = \frac{\delta_2}{2\pi} = \frac{45}{524288\pi^9} \approx 2.879 \times 10^{-9} \quad \text{[C].}$$

The *value* is an exact compiler number (only $c_3, \Omega_{\text{adm}}, B\gamma$; trace reading $\delta_2 = 4! \text{Tr}_{S^+}(X^2) c_3^8$, $\text{Tr} = 120 = 5!$) — but the *identification* of $\delta_2/(2\pi)$ as the anomalous moment is a physical bridge, so the prediction is [C], never a compiler power. [verification/v204_muon_seam_g2.py]

Data, honestly. Against the dispersive discrepancy $\Delta a_\mu = (2.49 \pm 0.48) \times 10^{-9}$ (Fermilab/BNL world average minus the WP2020 SM) this is 0.81σ . **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ($\Delta a_\mu \sim 1.5 \times 10^{-9}$); the TFPT value is *fixed* and would then sit $\sim 1.5\sigma$ high. A converged Δa_μ outside $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ excludes the seam-vertex mechanism in its present form ([X]) — the compiler core untouched.

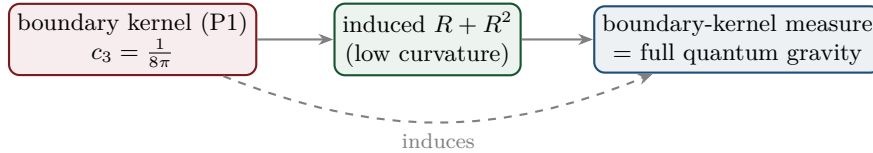
7 Full quantum gravity — induced from the boundary kernel

Framing (read this first): full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through v_{geo} and F_{transfer} and are falsifiable now (JUNO, CMB-S4, EDM). G_{net} certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation $c_3 = 1/(8\pi)$ is the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The R^2 /scalon sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an independent input*: fixing the physical Λ branch ($(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ relative to Λ , so G_N is read out *after* selecting the physical Λ branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [verification/v60_lambda_metrology_branch.py]) — a metrology readout, not a from-nothing prediction.

Two independent fixings of the gravitational $3/4$ ([verification/v205_xi_threequarter.py]). The induced-gravity coefficient the seam-determinant replica pins, $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$ ([verification/v152_norm_is_anchor.py]; the collar data exhibit this as the anchor — drifting μ_{lat} , no *derived* $\ln(m/\mu) = \frac{3}{4}$, v471), is reached a *second*, independent way: the old torsion-compression ansatz $\kappa^2 = \xi \varphi_0^{\text{ret}}/c_3^2$ with the Einstein limit $\kappa^2 = 8\pi G$ forces $\xi = c_3/\varphi_0^{\text{ret}}$, which at tree level $\varphi_0^{\text{ret}} \rightarrow 1/(6\pi)$ is *exactly* $3/4$ (with the retained seed 0.7483, -0.22%). So $3/4$ appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and $1/G$ stays the one declared anchor.

The Λ branch sharpens H_0 ([verification/v206_h0_lambda_branch.py]). Reading the seam vacuum number $\delta_\Sigma = \Omega_{\text{adm}} e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ (lowest admissible block $m_1 = \Omega_{\text{adm}} = 48$) through the flatness closure gives a TFPT Hubble value $H_0 = 66.5\text{--}67.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ($1.6\text{--}0.5\sigma$ below Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and H_0 uses the imported $\Omega_b, \omega_{\text{DM}}$, so it is a [C] readout; $|\log_{10} \delta_\Sigma| \approx 122.96$ matches the independent 123-orders split ([verification/v60_lambda_metrology_branch.py]), and $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4 = (8\pi)^2$ must be displayed, never silently interchanged.



Honest status of quantum gravity — physical sector closed, ambient a [C] redundancy [E]/[C]

“Full quantum gravity beyond the R^2 sector” is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action $\text{Tr } f(D/\Lambda)$ gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ($a_2 \sim -R/3$, $a_4|_{R^2} \sim R^2/72$), Einstein–Hilbert + R^2 *structurally* — the $R + R^2$ form is not postulated; the scalaron ratio $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ with the TFPT closure c_3^7 fixing the cutoff moment f_0 . *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([verification/v255_spectral_action_expansion.py]), and the cutoff moment f_0 is no longer a free scheme choice: identifying f with the *seam KMS weight* ($\beta=1$ by Tomita–Takesaki, $2\pi=1/(4c_3)$) gives $f_2/f_0 = 1$ exactly ([verification/v259_modular_cutoff_kappa.py]), so $\kappa = \sqrt{(f_2/f_0) c_{\text{PS}}/c_{\text{grav}}}$ collapses to a finite-triple trace ratio — the scalaron normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated, $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$, so the admissible sector keeps a strictly positive effective gap $\Delta_{\text{eff}} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the ambient, so under those assumptions the admissible-sector measure is closed. **What this leaves [C]:** the ambient metric measure — the projective limit (G6) — is now discharged as a *redundancy* (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann — the one open premise

being that the seam net condenses to the *holomorphic* $|\det K|=1$ point that selects $(E_8)_1$; the central charge $c=8$ alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample, v277/v281), a certification object rather than missing dynamics; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [verification/v36_spectral_action_g2.py] (iii) **The classical field equation is parameter-free** ([verification/v358_grav_entropy_equilibrium.py]). Beyond the $R+R^2$ action, the metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta \langle K \rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms: equilibrium gives the *linearised* Einstein equation $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed* — the four c_3 -roles (Unruh $1/(2\pi)=4c_3$, EH $1/(16\pi)=c_3/2$, Einstein $8\pi=1/c_3$, Bekenstein $1/4=1/|\mu_4|$) are mutually consistent, so *no free dimensionless Newton dial enters*. The *new* content is a coincidence: the first law needs the coefficient $2\pi/\eta$ (η the entropy density), the geometry gives $1/c_3 = |\mathbb{Z}_2|2\pi\chi(S^2)$, and these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — so the thermodynamic and geometric origins of c_3 are *one*, and c_3 is triply over-determined (anchor v23, geometry v58, thermodynamics v358). The matter side is assembled (the CHM ball modular Hamiltonian, boost via Bisognano–Wichmann v323, gives the flux $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$) and the entropy density is atom-fixed ($1/4=1/|\mu_4|$, central charge $c = g_{\text{car}} + N_{\text{fam}} = 8$). (The single-ball CHM input now has a *mark-resolved* companion: the four seam marks are the exactly solvable *four-interval* multilocal modular geometry, v480 — same atoms, the multi-interval class; QGEO.SYM.01 stays [O].) **Full covariant equation** ([verification/v359_grav_nonlinear_einstein.py]). Demanding stationarity at fixed *volume* (Jacobson 2015) brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab} ; trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in $d=4$) — the unique divergence-free metric tensor (Lovelock), so the equilibrium yields the *full* covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with $\nabla^a T_{ab}=0$ an output, and TFPT fixes *both* coefficients ($8\pi=1/c_3$ from (iii); Λ from $\alpha, \rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, v60). So B6's full covariant field equation is now parameter-free at the *local* level. **Residual:** only the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate for the missing action level — Bianconi's entropic action, Phys. Rev. D 111, 066001 (2025) — is quantified in v473: $\beta'_B = c_3/6$ pinned, R^2 gap $3(8\pi)^9$ pre-registered as kill test, typing unchanged) and the absolute scale v_{geo} remain [O] — the global ambient measure (C7=QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics; no free dimensionless parameter survives in the gravity coupling. So the metric sector now reads: $R+R^2$ action (i), gap-decoupled measure (ii), and a *parameter-free* full covariant classical field equation (iii); the ambient nonperturbative measure is a [C] redundancy, not an open structural hole. (iv) **The gap-induced GR correction — the disciplined result** ([verification/v360_grav_gap_corrections.py]). Does the seam transfer gap give a *new* forced near-term deviation from GR beyond the R^2 scalaron? Run with the v354/v355 discriminator, the honest answer is *no*: the leading higher-curvature term *is* the R^2 scalaron ($M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$), which is the next-order term of the same entanglement- equilibrium route (the Wald-entropy first law $\rightarrow f(R)=R+R^2$) and is *already* a live CMB-S4 kill test (n_s, r). A *distinct* gap-induced curvature² term enters only at the seam/Planck scale ($\Delta = 6 \ln \frac{3}{2}$ is a dimensionless rate, so the scale is ξv_{geo}), and a new dimensionless coefficient there is *not* atom-forced (the scalaron exponent $7=g+N-1$ and the gap exponent $6=2N_{\text{fam}}$ are different structures) — so it is [O] declined, not manufactured. The calculable GR deviation is the scalaron. (v) **Matter–gravity backreaction** ([verification/v361_grav_backreaction.py]). Feeding the carrier stress tensor into the parameter-free equation: the carrier's gravitational anomaly is forced, $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ (the chiral central charge of the 16-Majorana content, the same c_- as (iii)); and the matter vacuum-energy backreaction is *finite* — gap-suppressed (the Decoupling Theorem) to the forced $\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), *not* the M_{Pl}^4 catastrophe — so the loop (the equation $+ \Lambda +$ the carrier vacuum) closes. Beyond $c_-=8$ and Λ there is [O] no new atom-forced gravitational read-out of individual SM quantum numbers (the backreaction is GR with the SM content); the discipline declines manufacturing one.

Update (the QG.AMB.01 roadmap, consolidated — [verification/v275_qgamb_roadmap.py]). The gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling, $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 2.248 c_3^2 = 1.648 > 0$) makes every low-energy readout independent of the ambient measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c=248/31=8$, v77), now *closed modulo cited theorems* on the MMST route SEAM.EQUIV.MMST.01 (lattice v367/v368 + the S_3 stack v376–v379, Lean FORM.SEAM.MMST.01; the parent SEAM.EQUIV.01 stays [O]); the *perturbative* 4d layer is now in hand (v269/v271/v273). [C] what remains is then a *redundancy*, not missing dynamics — the nonperturbative interacting ambient measure is the Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), *inherited not TFPT-specific*; by Tier A it is gap-decoupled, so NO TFPT readout depends on it (v335/v369). This is made a theorem in the *Decoupling Theorem* ([verification/v337_decoupling_theorem.py]): every dimensionless readout

factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, whose finite susceptibility $\chi = 729/665$ gap-suppresses the ambient contribution (margin $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$), and the negative control (gap closed $\Rightarrow \chi$ diverges) makes it non-vacuous — so TFPT provably does not *need* the ambient measure. It is therefore reclassified from “TFPT structural frontier” to a **[C]** *redundancy* (a certification object, conditional on SEAM.EQUIV.01+Bisognano–Wichmann, v369): its obstruction is characterised (v332) and conditionally resolved by the GHP contour rotation + IDG dressing (v334); not a TFPT structural hole (QGAMB.ROADMAP.01). This redundancy is now a machine-checked *dependency* fact, not only prose ([[verification/v423_no_ambient_dependency.py](#)], QGAMB.NODEP.01): QG.AMB.01 is a certification *sink* — every claim that lists it is a gravity-gate/QG row, never a frozen prediction — and it is *not a value source* — it has no producing script, while α^{-1} , the mixing angles and the cosmological readouts are each *computed* by a compiler script (v3/v9/v268/v7), none of them an ambient-measure script. The single transitive link is architectural, not numerical: of the frozen readouts only θ_{13} references QG.AMB.01, through the 4d-QFT contract spine (PS.DIRAC $\rightarrow$ CONTRACT.QFT4D $\rightarrow$ QG.G2 $\rightarrow$ GATE.METRIC) — a coherence edge from living inside the unified spectral-action construction, not an input to its value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ (v268). Honest scope: this kills QG.AMB.01 specifically; it does *not* discharge SEAM.EQUIV.01, on which the readouts still depend.

Tier B made concrete ([[verification/v277_seam_calderon_e8_match.py](#)]). The seam-Calderón $\rightarrow (E_8)_1$ identification is reduced to a *single* bit. **[E]** the full $(E_8)_1$ matching target is computed and matches: $c=8$ (Sugawara 248/31), $\det \text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248 = \dim E_8$ weight-1 currents. **[E]** the discriminator is *holomorphy*: the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det \text{Cartan}=4$ (four primaries, non-holomorphic) — same c , different det, so the one discriminating condition is $|\det K|=1$. **[C]** the seam side matches via the flat $\tau=i$ pillowcase (v276) and the Kummer/K3 (v260). **[C]** the residual is exactly that single holomorphy bit — now *closed modulo cited theorems* on the MMST route SEAM.EQUIV.MMST.01 (lattice v367/v368 + the S_3 stack v376–v379, Lean FORM.SEAM.MMST.01; cited continuum existence **[O]**, v336) (QGAMB.TIERB.01).

The modular-data view ([[verification/v281_holomorphy_modular_data.py](#)]). The Tier-B bit is the anyon count: $\#\text{anyons} = |\det \text{Gram}|$ ($E_8 \rightarrow 1$ holomorphic, $SO(16) \rightarrow 4$), the condensation tower $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$ each step a Lagrangian Z_2 boson, and Gauss–Milgram returns $c=8$ — so the open bit is simply “does the seam net condense to $\det K=1$ ” (QGAMB.MODULAR.01).

The live kill-test scorecard ([[verification/v283_live_killtest_scorecard.py](#)]). The recent round’s computable predictions confronted with current data: $\sin^2 \theta_{12}$ ($+0.31\sigma$), $\sin^2 \theta_{13}$ ($+1.45\sigma$), $J_{\text{PMNS}} = -0.0297$, Σm_ν floor 0.0586 eV, proton decay $\tau_p = 1.55 \times 10^{35}$ yr — all consistent at $< 2\sigma$, with two sharp near-term kill tests (Σm_ν : DESI/CMB-S4; proton decay: Hyper-K) (PRED.LIVE.01). Two *named post-hoc* **[O]** candidates sit next to this board without touching the frozen record: the third-generation ladder-base pattern (the three $\sim 2\sigma$ mixing tensions $\theta_{13}/\theta_{23}/|V_{cb}|$ as one common $-\varphi_0^{\text{ret}}$ suppression, FLAV.THIRDDGEN.PATTERN.01, v467) and the splitting-ratio relation $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$ at -0.19σ (FLAV.DM2RATIO.01, v468) — both pre-registered with symmetric kills (JUNO / Belle II), see document 2.

Update (QBL chain, v108–v113): the boundary-net residual now carries *double* weight — the carrier choice $g = 5$ has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank = c : 5 carrier block, 8 seam hull) is the defining datum of the free $c = 8$ net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection **[E]** (see `tfpt_1` and `tfpt_research_contracts`). [[verification/v113_quasifree_kernel.py](#)]

Update (free-bulk premise discharged, v160–v165): that “free $c=8$ net” premise is no longer free-standing. It is a fixed-point theorem (quasi-free $\Rightarrow$ all higher cumulants $\kappa_{2n}=0$, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), forced by μ_4 -equivariance (v162); and the residual factors into the already-open A_2 net-existence (an assembly of established net constructions, **[C]**) and R_1 is the seam-realisation keystone SEAM.EQUIV.01, with the irreducible core $\{\pi, v_{\text{geo}}\}$ a theorem (v165). So the metric-sector gate carries *zero new* independent open content. [[verification/v165_premise_a_closure.py](#)]

The deduction below the premise is Lean-formalised (FORM.QGEO.02/FORM.QGEO.01). After the residual reduces to the single seam-realisation postulate QGEO.SYM.01 (the carrier μ_4 clock is the conformal deck of the seam; `tfpt_research_contracts`), its geometric normal form (the UNIFORM node: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, $\text{orbit} \rightarrow \mu_4$, multiplier order-4 $\Leftrightarrow \zeta = \pm i$) and the conditional im-

plication $\text{mark-local} \Rightarrow \omega \circ \rho = \omega$ are machine-proved in the carrier-rigidity Lean project (AUDIT: PASS, only the three standard kernel axioms; mirrors `[verification/v177_seam_marking_kernel.py]` and `[verification/v210_mark_local_dtn.py]`). So everything *downstream* of the postulate is [E]; QGEO.SYM.01 *itself* is the downstream face of the keystone SEAM.EQUIV.01, which is now *closed modulo cited theorems on its MMST route* SEAM.EQUIV.MMST.01 (lattice + S_3 stack, Lean FORM.SEAM.MMST.01; only the cited continuum existence stays [O] — its extension leg since re-founded on peer-reviewed crossed-product theorems, the realisation input reduced to invariant level $R1'$, v469; the twistor route SEAM.EQUIV.TWISTOR.01 and the unconditional parent stay [O]).

A second, independent route (`[verification/v207_asymptotic_safety.py]`, [O]). Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point — $\partial_t g = (2 + \eta_R)g$ is UV-repulsive at $g \rightarrow 0$ and damped at large g by the positive-definite torsion K^2 term, so $2 + \eta_R = 0$ has a root $g_* > 0$ (intermediate value), with the scale-invariant axion coupling fixed and small, $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$. This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close G_{metric} ; the primary, machine-checked route stays the seam-net. A third, quantitative approach to the keystone — the celestial and twistor continuum route, now named SEAM.EQUIV.TWISTOR.01 (contract CELEST.SEAM.01, WP1–WP5d, both WP5d stages, plus the WP5e $\alpha, \beta, \gamma, \delta_1, \delta_2, \varepsilon_2$ and ε_1 stages and the M1–M3 back-reaction milestones executed: the level-2 null ideal derived, its deleting operator explicit, the GNS limit state constructed, the two-interval index chain measured, split + strong additivity witnessed on the lattice (all three KLM ingredients of complete rationality; the continuum uplift is Xu’s theorem, cited not claimed), the $q^{-1/3}$ prefactor + level $k=1$ pinned on the CFT side, the equivariant anomaly ledger exact on the twistor side (index bridge, -78 defect, rigid $32T_3$ residual, the honest a_0 refutation), the sphere-axion exchange sub-branch closed with a certificate, the collapse confirmed full-tensorially with the one cubic d -symbol door open (its $c_d = 1920 = |W(D_5)|$ fingerprint honestly typed look-elsewhere-loaded and convention-contingent, the stable core $c_d = 32 \times 60$, v513), “one level” lifted to a theorem of clock invariance with the sector counter pinning $k = 1$ (v508/v509/v511/v513), the $O(-2)$ bulk-axion slot built with $\tilde{\lambda} = 6$ triply pinned (v514), the back-reacted Ω_N closed-form with $(2\pi i)^2$ -integral lockstep periods and the lens-forced source charge $4 = |\mu_4|$ (v515), the twisted KS measure on the declared completion reading cancelling the $32T_3$ residual and supplying the $\psi = 64$ slice without the cubic d -channel (v516), the a_0 uplift coupled to the centre count on four scales, monodromy i (v517), the δ_1 chain decided — kill under the derived measure, the declared/derived measure tension stated honestly (v518) — the measure question since decided at probe level for the declared reading (single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4, v520), and the w_m normalisation since derived constructively at probe level ($1/\det_j$ = the Atiyah–Bott/zeta-determinant fixed-point factor, three independent sources, v523); WP5e proper [O], the global BCOV quantisation — narrowed to the global BCOV integral beyond the fibre zero-mode factor — the single remaining milestone) — is synthesised as a dedicated section in `tfpt_3_e8_audit_bootstrap`; it does not close SEAM.EQUIV.01 either.

The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]

The single dimensionful input (v_{geo} = the reduced Planck mass M_{Pl}) is not a hidden gap but the *unit*, and it is over-determined (`[verification/v274_scale_overdetermination.py]`). [E] every dimensionful output is a pure number $\times M_{\text{Pl}}^d$: $\rho_\Lambda/M_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$, $M_{\text{scal}}/M_{\text{Pl}} = c_3^{7/2}$, the whole spectrum — the absolute ν -scale (v272) and v_{geo} are the *same* anchor. [C] **two independent routes agree**: gravity gives $M_{\text{Pl}} = (8\pi G)^{-1/2} = 2.4353 \times 10^{18}$ GeV, and inverting the compiler’s dark-energy prediction with the measured $\rho_\Lambda^{1/4} = 2.24$ meV gives $M_{\text{Pl}} = 2.438 \times 10^{18}$ GeV — the same Planck mass to 0.11% (*conditional* on the Λ -branch readout LAMBDA.BRANCH.01, [C]; the gravity route is unconditional — red-team `[verification/redteam/rt_F_qft4d.py]`). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal $c=8$ chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (SCALE.OVERDET.01).

Summary

Item	Status	Honest handle / what is missing
η_B	[C]	downstream readout = 6.1×10^{-10} from closed $\Omega_b h^2$; leptogenesis interface, cleanest scenario $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$, $\tilde{m}_1 = m_3/A_\Lambda$ (shared decouple, no seesaw scale)
m_p/m_e	[C]	cross-sector (QCD scale / closed m_e); typed runnable solver with $\pm 7.5\%$ uncertainty budget (v374), not a compiler power
Koide Q	[C]	source ratio $Q = 0.664$ near $2/3$; the source→pole factor $53/54$ is an <i>exact operator readout</i> of the missing $F=R+Q$ Sheet-Diamond corner [E]; the physical pole transfer (F_{pole}) stays [C]
dark matter	[C]	determinant-line axion: $f_a = M_{\text{scal}}/128$, $m_a \approx 23.8 \mu\text{eV}$. <i>Post-inflationary</i> ($H_{\text{inf}}/f_a \approx 66$, isocurvature): the misalignment angle is a horizon average (<i>not</i> observable), the relic follows from f_a (in the post-inflationary window), domain-wall-safe via $N_{\text{DW}} = C = 1$ (v470/v472); the $3\pi/5$ spine value is retained as a structural rational, not a relic prediction — branch-level kill test, not a theory closure
muon a_μ	[C]	seam vertex $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) = 2.879 \times 10^{-9}$; exact <i>value</i> , but the vertex projection is the [C] bridge (0.81σ dispersive, $\sim 1.5\sigma$ lattice)
quantum gravity	[E]/[C]	local field eq. parameter-free ($G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$, v358/v359); $R + R^2$ heat-kernel grounded (G2); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure (G6) a [C] redundancy (v369)

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status_ledger.csv.

Net (honest). Two of the five have a genuine quantitative handle that already lands on the data ($\eta_B = 6.1 \times 10^{-10}$ via the closed Ω_b ; the axion DM candidate with a closed misalignment angle). One is a computed *near-miss* reported without tuning (Koide 0.664 vs $2/3$). One is a typed runnable transfer (m_p/m_e as a cross-sector ratio, v374); quantum gravity has been sharpened — its local field equation is now *parameter-free* (the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ from fixed-volume entanglement equilibrium, v358/v359), its $R + R^2$ form heat-kernel grounded (G2), its physical sector gap-decoupled (G5), and *perturbative* graviton unitarity established (the Stelle ghost a truncation artefact, v304/v370/v380; the entire-form-factor graviton-exchange amplitude $e^{-p^2/M^2}/p^2$ is finite, UV-softened and tree-unitary, [verification/v386_grav_amplitude.py], and UV-finite at every loop order by power counting — the super-polynomial falloff drives the superficial degree of divergence to $-\infty$, regulated by the seam scale M_{scal} , [verification/v389_grav_loop_finiteness.py], with full loop *unitarity* of the nonlocal form factor honestly the still-open point; the nonlocal action's IR projection *is* the atom-scaled $R + R^2$, $M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$, so the local readout is its shadow, not a truncation with sawn-off legs, [verification/v400_grav_nonlocal_action.py]), leaving the ambient projective *measure* (G6) as a [C] redundancy (v369), not an open hole. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

The F_{transfer} functor as typed, runnable solvers. The four frontier transfers are one functor — compiler *source* → measured observable via cited standard physics — now promoted from contracts to runnable solvers with kill tests, each [C] (never a compiler power). *Koide* ([verification/v371_ftransfer_pole.py]): a clean *negative* — standard QED running cannot be the source→pole transfer (it pushes Q *above* $2/3$; the required ϵ is negative while QED's is positive), so the transfer is the non-QED $53/54$ operator / $(2/3)^6$ Moebius generator (v183/v82). η_B ([verification/v372_ftransfer_boltzmann.py]): the integrated BDP Boltzmann ODE at the frozen $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ gives η_B within a factor ~ 1.1 of 6.1×10^{-10} with no M_R dial. *Axion* ([verification/v373_ftransfer_relic.py]): the finite- T misalignment ODE *decides* the branch — the frozen spine angle $3\pi/5$ lands $\Omega_a h^2$ on Ω_{DM} (in band) while the hilltop 170° over-produces ($\sim 5.4 \times$);

in the natural post-inflationary scenario ($H_{\text{inf}}/f_a \approx 66$) the angle is instead a horizon average and the relic follows from f_a with $N_{\text{DW}}=1$. m_p/m_e ([`verification/v374_fttransfer_qcd.py`]): the observed 1836.15 sits inside the propagated $\pm 7.5\%$ band of the two declared externals ($\alpha_s(M_Z)$, C_p), with lattice tightening as the kill. Each remains [C] — a transfer target, not a forced compiler number.

Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is $\Omega_b = (1 - \frac{1}{4\pi})\varphi_0^{\text{ret}}$ (whence the η_B order) in `verification/v7_gravity_cosmo.py`. The $R + R^2$ structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); the proton/electron ratio is now a typed runnable [C] transfer (`v374`) and the ambient projective limit (G6) a [C] redundancy (`v369`) — both conditional, not bare [O]. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, [arXiv:2107.13834](https://arxiv.org/abs/2107.13834), [doi:10.1007/s00220-022-04521-8](https://doi.org/10.1007/s00220-022-04521-8). (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur $\rightarrow$ Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token “MMST” and the legacy claim-ids SEAM.MMST.* / filenames v*_mmst_* denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, [arXiv:2010.11121](https://arxiv.org/abs/2010.11121), [doi:10.1007/s00220-021-04152-5](https://doi.org/10.1007/s00220-021-04152-5); and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)

- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, [arXiv:2407.18222](#) (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, [arXiv:2506.01008](#) (2025); *Representations of conformal nets associated with infinite-dimensional groups*, [arXiv:2508.07109](#) (2025).
- P. Naaijken, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, [arXiv:2605.10693](#).

Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), *Ann. of Math.* **36** (1935) 527–606.

Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, *Invent. Math.* **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, *Ann. of Math. Studies* **61**, Princeton Univ. Press (1968).

Spectral geometry, gravity and thermal time.

- A. H. Chamseddine, A. Connes, *The spectral action principle*, *Comm. Math. Phys.* **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, *Class. Quantum Grav.* **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, *Phys. Rev. D* **48** (1993) R3427–R3431.
- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, *JHEP* **05** (2011) 036, [arXiv:1102.0440](#).